Project: Worked examples
Title: Beam and 2D frame analysis
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates reactions, member forces and deflections of a linear elastic plane frame or continuous beam by the direct stiffness method. Global x is right and y up; bending is about the section y-y axis, anticlockwise positive.

WHAT IT CALCULATES	RELATION
Member stiffness	three freedoms per node, Euler–Bernoulli members with axial EA/L and bending EI/L³ terms
End releases	pins are taken out by static condensation; supports remove freedoms or add spring stiffness
Span loads	they become equivalent nodal loads, the exact fixed-end forces
Nodal displacements	K u = F, solved by banded LDL▒ after a reverse Cuthill–McKee ordering; a non-positive pivot is reported as a mechanism
Internal forces	they follow from the member end actions and the Macaulay functions of the span loads
Member extremes	found at stations along each member, plus the zero-shear and zero-slope points found by bisection

Basis: Plane frame analysis using the stiffness method[1](Ch. 16, 9th ed.); Shear, moment, slope and deflection formulas for elastic straight beams[2](Table 8.1, 8th ed.); Combinations of actions (user-entered factors)[3](cl. 6.4.3).

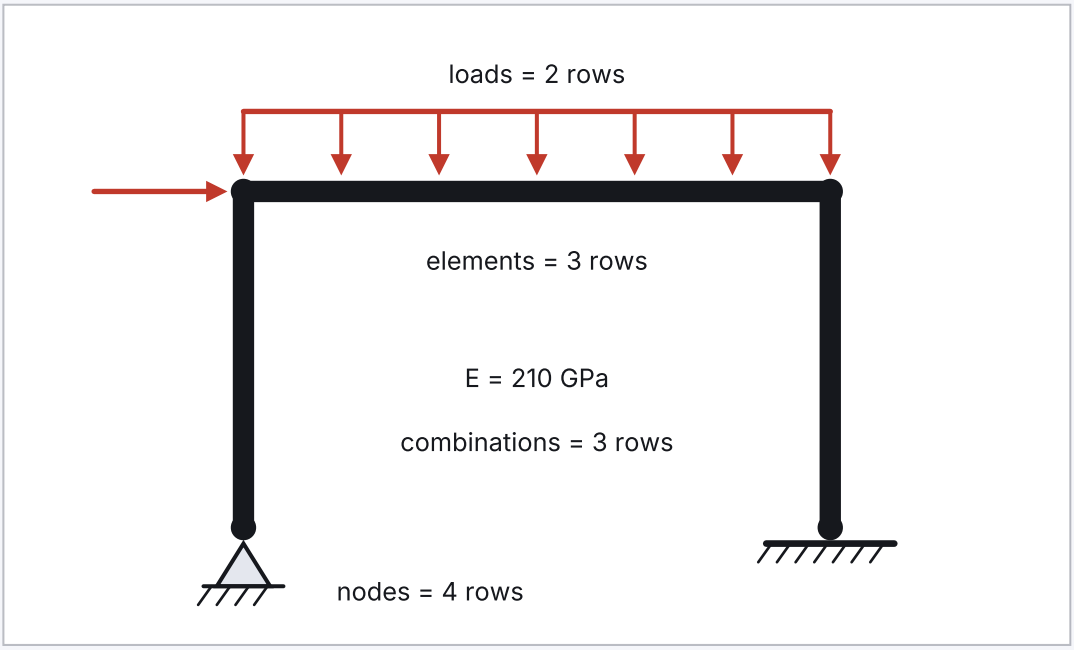


Figure 1 — A plane frame (a portal drawn representatively): the nodes, elements, supports and loads as listed. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
nodes	Nodes	4 row(s)		drawing: DRG-PF-010 rev B	any
elements	Elements	3 row(s)		reference: UC 203×203×46 columns, UB 305×165×40 rafter (Blue Book)	any
loads	Loads	2 row(s)		spec: Load schedule LS-02: roof 15 kN/m; wind 24 kN at eaves	any
n_{div}	Stations per member	16		assumption: 16 stations per member.	4 … 100
combinations	Load combinations (optional; blank = each case alone)	3 row(s)		assumption: Characteristic combinations for the worked example.	any
report	Combination to report in detail (optional; blank = the first)	G+W		assumption: Gravity plus wind reported.	any
E	Default Young's modulus	210	GPa	spec: BS EN 1993-1-1 3.2.6	1 … 500 GPa

TABLE ROW	VALUES
nodes[1]	id = 1; x = 0 m; y = 0 m; support = pin
nodes[2]	id = 2; x = 0 m; y = 5 m; support = free
nodes[3]	id = 3; x = 8 m; y = 5 m; support = free
nodes[4]	id = 4; x = 8 m; y = 0 m; support = pin
elements[1]	id = 1; i = 1; j = 2; A = 58.7 cm²; I = 4568 cm⁴; rel_i = no; rel_j = no
elements[2]	id = 2; i = 2; j = 3; A = 51.3 cm²; I = 8503 cm⁴; rel_i = no; rel_j = no
elements[3]	id = 3; i = 3; j = 4; A = 58.7 cm²; I = 4568 cm⁴; rel_i = no; rel_j = no
loads[1]	case = 1; kind = udl; target = 2; a = 0 m; b = 8 m; w1 = -15 kN/m; w2 = 0 kN/m; F_x = 0 kN; F_y = 0 kN; M = 0 kN·m
loads[2]	case = 2; kind = nodal; target = 2; a = 0 m; b = 0 m; w1 = 0 kN/m; w2 = 0 kN/m; F_x = 24 kN; F_y = 0 kN; M = 0 kN·m
combinations[1]	name = G+W; case = 1; factor = 1
combinations[2]	name = G+W; case = 2; factor = 1
combinations[3]	name = G; case = 1; factor = 1

3. Assumptions

A1. The analysis is linear elastic and first order, with small displacements. (Euler–Bernoulli direct stiffness.)
A2. Members have no shear deformation, and there are no P–Δ effects or buckling. (Euler–Bernoulli direct stiffness.)

A3. Members are prismatic between nodes and connect rigidly unless a release is stated. (Static condensation of the released rotation.)
A4. A released member end transmits no moment. (Static condensation of the released rotation.)
A5. Distributed loads act perpendicular to the member unless marked as gravity loads. (Load table convention.)
A6. Gravity loads act in global −y per metre of member length. (Load table convention.)
A7. n_{div} (Stations per member) is taken as 16. (16 stations per member.)
A8. combinations (Load combinations (optional; blank = each case alone)) is taken as 3 row(s). (Characteristic combinations for the worked example.)
A9. report (Combination to report in detail (optional; blank = the first)) is taken as G+W. (Gravity plus wind reported.)

4. Working

[4] n_{nodes} = nodes
= 4
= 4
Nodes in the model

[5] $n_{elements}$ = elements
= 3
= 3
Members in the model

[6] n_{comb} = combinations
= G+W, G
= 2
Load combinations analysed

[7] comb = reported combination
= G+W
= "G+W"
The combination detailed below

[8] $R1_{x}$ = reaction at node 1, x
= Σ member end forces − applied nodal loads
= -2.9968 kN
Horizontal reaction at node 1

[9] $R1_{y}$ = reaction at node 1, y
= Σ member end forces − applied nodal loads
= 45 kN
Vertical reaction at node 1

[10] $R1_{m}$ = reaction at node 1, moment
= Σ member end moments − applied nodal couple
= -1.4552e-14 kN·m
Moment reaction at node 1

[11] $R1_{x,abs}$ = $|R_x|$ at node 1
= |-2.9968| kN
= 2.9968 kN
Magnitude of the horizontal reaction at node 1

[12] $R1_{m,abs}$ = |M| at node 1
= |-1.4552e-14| kN·m
= 0 kN·m
Magnitude of the moment reaction at node 1

[13] $R4_{x}$ = reaction at node 4, x
= Σ member end forces − applied nodal loads
= -21.003 kN
Horizontal reaction at node 4

[14] $R4_y$ = reaction at node 4, y
= Σ member end forces − applied nodal loads
= 75 kN
Vertical reaction at node 4
[15] $R4_m$ = reaction at node 4, moment
= Σ member end moments − applied nodal couple
= -2.9104e-14 kN·m
Moment reaction at node 4
[16] $R4_{x,abs}$ = $|R_x|$ at node 4
= |-21.003| kN
= 21.003 kN
Magnitude of the horizontal reaction at node 4
[17] $R4_{m,abs}$ = |M| at node 4
= |-2.9104e-14| kN·m
= 0 kN·m
Magnitude of the moment reaction at node 4
[18] $sum_{R,y}$ = Σ R_y
= all supports
= 120 kN
Total vertical reaction (equilibrium check against the applied load)
[19] $sum_{R,x}$ = Σ R_x
= all supports
= -24 kN
Total horizontal reaction
[20] $M_{max,e1}$ = member 1: max sagging M
= at x = 5 m from node 1
= 14.984 kN·m
Member 1 maximum moment
[21] $M_{min,e1}$ = member 1: min (hogging) M
= at x = 0 m
= 0 kN·m
Member 1 minimum moment
[22] $M_{abs,e1}$ = member 1: max |M|
= max(|14.984|, |0|) kN·m
= 14.984 kN·m
Member 1 greatest moment magnitude (sagging or hogging)
[23] V_{e1} = member 1: max |V|
= 2.9968 kN
= 2.9968 kN
Member 1 maximum shear
[24] N_{e1} = member 1: max |N| (tension +)
= -45 kN
= -45 kN
Member 1 axial force of greatest magnitude
[25] d_{e1} = member 1: max |deflection|
= at x = 5 m
= 74.678 mm
Member 1 maximum transverse deflection
[26] $M_{max,e2}$ = member 2: max sagging M
= at x = 3 m from node 2
= 82.484 kN·m
Member 2 maximum moment

[27] $M_{min,e2}$ = member 2: min (hogging) M
= at x = 8 m
= -105.02 kN·m
Member 2 minimum moment
[28] $M_{abs,e2}$ = member 2: max |M|
= max(|82.484|, |-105.02|) kN·m
= 105.02 kN·m
Member 2 greatest moment magnitude (sagging or hogging)
[29] V_{e2} = member 2: max |V|
= 75 kN
= 75 kN
Member 2 maximum shear
[30] N_{e2} = member 2: max |N| (tension +)
= -21.003 kN
= -21.003 kN
Member 2 axial force of greatest magnitude
[31] d_{e2} = member 2: max |deflection|
= at x = 3.4916 m
= 25.45 mm
Member 2 maximum transverse deflection
[32] $M_{max,e3}$ = member 3: max sagging M
= at x = 5 m from node 3
= 0 kN·m
Member 3 maximum moment
[33] $M_{min,e3}$ = member 3: min (hogging) M
= at x = 0 m
= -105.02 kN·m
Member 3 minimum moment
[34] $M_{abs,e3}$ = member 3: max |M|
= max(|0|, |-105.02|) kN·m
= 105.02 kN·m
Member 3 greatest moment magnitude (sagging or hogging)
[35] V_{e3} = member 3: max |V|
= 21.003 kN
= 21.003 kN
Member 3 maximum shear
[36] N_{e3} = member 3: max |N| (tension +)
= -75 kN
= -75 kN
Member 3 axial force of greatest magnitude
[37] d_{e3} = member 3: max |deflection|
= at x = 0.31515 m
= 75.042 mm
Member 3 maximum transverse deflection
[38] M_{max} = max sagging moment
= member 2 at x = 3 m
= 82.484 kN·m
Maximum moment in the reported combination
[39] x_{Mmax} = position of M_{max}
= from node i of member 2
= 3 m
Position of the maximum moment

[40] M_{min} = min (hogging) moment
= member 3 at x = 0 m
= -105.02 kN·m
Minimum moment in the reported combination
[41] V_{max} = max |V|
= member 2
= 75 kN
Maximum shear in the reported combination
[42] N_{max} = max tension
= member 2
= -21.003 kN
Maximum axial tension
[43] N_{min} = max compression (negative)
= member 3
= -75 kN
Maximum axial compression
[44] d_{max} = max |deflection|
= member 3 at x = 0.31515 m
= 75.042 mm
Maximum transverse deflection
[45] $M_{max,env}$ = envelope max M
= G+W, member 2
= 82.484 kN·m
Maximum moment over all combinations
[46] $M_{min,env}$ = envelope min M
= G+W, member 3
= -105.02 kN·m
Minimum moment over all combinations
[47] $V_{max,env}$ = envelope max |V|
= G+W, member 2
= 75 kN
Maximum shear over all combinations
[48] $d_{max,env}$ = envelope max |δ|
= G+W, member 3
= 75.042 mm
Maximum deflection over all combinations

5. Results

SYMBOL	VALUE	DESCRIPTION
$R1_x$	-2.997 kN	R_x node 1
$R1_y$	45 kN	R_y node 1
$R1_m$	0 kN·m	M node 1
$R1_{x,abs}$	2.997 kN	$\|R_x\|$ node 1
$R1_{m,abs}$	0 kN·m	|M| node 1
N_{e1}	-45 kN	N, member 1
V_{e1}	2.997 kN	|V| max, member 1
$M_{abs,e1}$	14.98 kN·m	|M| max, member 1
$M_{max,e1}$	14.98 kN·m	M max, member 1
$M_{min,e1}$	0 kN·m	M min, member 1

SYMBOL	VALUE	DESCRIPTION
V_{e2}	75 kN	|V| max, member 2
$M_{abs,e2}$	105.02 kN·m	|M| max, member 2
$sum_{R,y}$	120 kN	Σ R_y
$sum_{R,x}$	-24 kN	Σ R_x
M_{max}	82.48 kN·m	M max
M_{min}	-105.02 kN·m	M min
V_{max}	75 kN	|V| max
N_{max}	-21.00 kN	N max (tension)
N_{min}	-75 kN	N min (compression)
d_{max}	75.04 mm	δ max
$R4_x$	-21.00 kN	R_x node 4
$R4_y$	75 kN	R_y node 4
$R4_m$	0 kN·m	M node 4
$R4_{x,abs}$	21.00 kN	|R_x| node 4
$R4_{m,abs}$	0 kN·m	|M| node 4
d_{e1}	74.68 mm	δ max, member 1
$M_{max,e2}$	82.48 kN·m	M max, member 2
$M_{min,e2}$	-105.02 kN·m	M min, member 2
N_{e2}	-21.00 kN	N, member 2
d_{e2}	25.45 mm	δ max, member 2
$M_{max,e3}$	0 kN·m	M max, member 3
$M_{min,e3}$	-105.02 kN·m	M min, member 3
$M_{abs,e3}$	105.02 kN·m	|M| max, member 3
V_{e3}	21.00 kN	|V| max, member 3
N_{e3}	-75 kN	N, member 3
d_{e3}	75.04 mm	δ max, member 3
x_{Mmax}	3 m	x of M max
$M_{max,env}$	82.48 kN·m	M max, envelope
$M_{min,env}$	-105.02 kN·m	M min, envelope
$V_{max,env}$	75 kN	|V| max, envelope
$d_{max,env}$	75.04 mm	δ max, envelope

6. Checks (governing first)

None.

7. Applicability

None.

8. Verification

```
Template struct.frame-2d@1.5.0 Content hash 563dd9a0eb5dfb9f Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Hibbeler RC. Structural Analysis. 10th ed. Pearson; 2018.
2. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
3. British Standards Institution. Eurocode: Basis of Structural Design. BS EN 1990. BSI.